

Aspect-Based Sentiment Analysis of Code-Mixed Data Using a Transliteration-Aware BiLSTM-Attention Model

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Abstract

Sentiment analysis is a key task in Natural Language Processing (NLP) that focuses on identifying and classifying opinions expressed in text into sentiment categories such as positive, negative, or neutral. With the exponential growth of social media, online review, large volumes of user-generated content are produced daily, making automated sentiment understanding essential for applications like opinion mining and recommendation systems [1, 2, 3]. However, social media, online reviews text is often noisy and informal, containing slang, emojis, and multilingual expressions, which makes accurate sentiment classification challenging.

This research focuses on Aspect-Based Sentiment Analysis (ABSA) of Hinglish code-mixed text, where Hindi and English are mixed in Roman script. The proposed approach incorporates a transliteration-aware preprocessing module to normalize inconsistent Romanized Hindi words, followed by embedding generation and a Bidirectional Long Short-Term Memory (BiLSTM) network to capture contextual dependencies [4]. An attention mechanism is used to focus on sentiment-relevant words at the aspect level, improving classification performance for positive, negative, and neutral sentiments.[5].

1 Introduction

Natural Language Processing (NLP) is a subfield of artificial intelligence that enables machines to understand, interpret, and generate human language in a meaningful way [3]. It combines computational linguistics, machine learning, and statistical methods to process and analyze large-scale textual data. NLP plays a crucial role in applications such as machine translation, question answering systems, chatbots,

and information retrieval, where understanding human language is essential [1, 2]. With the rapid expansion of digital communication platforms, NLP has become increasingly important for extracting useful insights from unstructured text data such as social media, online reviews, posts, product reviews, and online discussions.

Despite significant advancements, NLP systems face multiple challenges when dealing with real-world textual data. Social media, online reviews content is highly unstructured and often includes slang, abbreviations, emojis, spelling variations, and grammatical inconsistencies. These factors make it difficult for traditional models to accurately interpret meaning. Additionally, sarcasm, ambiguity, and context-dependent expressions further complicate language understanding. Multilingual and code-mixed text adds another layer of complexity, where multiple languages appear within the same sentence, often without consistent grammar or structure [6, 7].

To address these challenges, this work focuses on Aspect-Based Sentiment Analysis (ABSA), which aims to determine sentiment polarity with respect to specific aspects mentioned in a sentence. Unlike general sentiment analysis, ABSA provides fine-grained opinion mining by associating sentiment with particular entities or features. This makes it more suitable for analyzing real-world user feedback, where a single sentence may contain multiple opinions about different aspects [8, 9].

In this research, we specifically target Hinglish code-mixed text, where Hindi and English are mixed in Roman script. Such data is widely observed on online reviews and presents significant challenges due to inconsistent transliterations and non-standard spellings. For example, the same Hindi word may appear in multiple forms, such as “acha”, “achha”, or “accha”, leading to vocabulary sparsity and ambiguity.

To handle this issue, we propose a transliteration-aware preprocessing module that normalizes noisy Romanized Hindi words into a consistent form before further processing. This step reduces lexical variation

and improves the quality of input representations for downstream learning models.

Finally, the processed text is analyzed using a Bidirectional Long Short-Term Memory (BiLSTM) network to capture contextual dependencies in both forward and backward directions. An attention mechanism is integrated on top of BiLSTM to focus on sentiment-relevant words associated with specific aspects. This combination enables the model to effectively learn fine-grained sentiment patterns in Hinglish code-mixed data and improve classification performance for positive, negative, and neutral sentiment classes [10, 11].

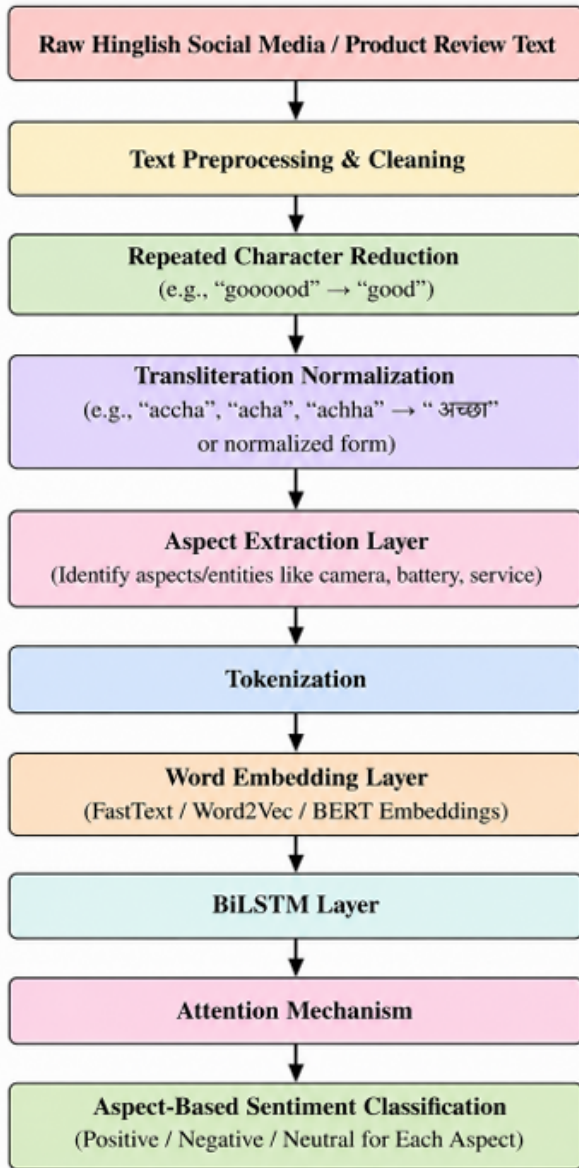


Figure 1: Proposed Framework for Sentiment Analysis

2 Problem Statement

Designing a transliteration-aware and context-sensitive ABSA framework that normalizes lexical variations and accurately predicts aspect-level sentiment in code-mixed text.

3 Literature Review

Aspect-Based Sentiment Analysis (ABSA) has evolved from document-level sentiment classification toward fine-grained identification of aspects and their associated opinions. Recent research has explored deep learning, recurrent, attention-based, and transformer architectures for ABSA, while highlighting challenges associated with noisy, informal, and multilingual text [8, 9]. Attention-based LSTM and BiLSTM architectures have demonstrated the importance of contextual information for aspect-level sentiment classification [12, 5].

With the increasing prevalence of code-mixed text, researchers have focused on handling linguistic noise, spelling variations, and transliteration. Studies on code-mixed reviews have explored normalization, BiLSTM, CNN-attention, and pretrained transformer-based approaches [13, 14, 15, 16]. More recent work has extended ABSA to e-commerce and Hinglish datasets, demonstrating the effectiveness of transformer-based and multilingual models [17, 18]. However, naturally occurring Hinglish reviews continue to present challenges due to transliteration, spelling variations, code-mixing, and informal expressions.

3.1 Research Gaps

Despite significant progress in deep learning-based ABSA, several research gaps remain in the analysis of naturally occurring Hinglish e-commerce reviews. Existing studies have primarily focused on standard English reviews or code-mixed social media datasets, with comparatively limited attention to naturally collected Hinglish reviews in the e-commerce domain [16, 17, 18].

- **Limited Handling of Transliteration and Spelling Variations:** Romanized Hindi does not follow a standardized writing convention, resulting in multiple spellings of the same word. Existing approaches provide limited explicit mechanisms for handling such transliteration and spelling variations, which can lead to vocabulary fragmentation and reduced robustness.
- **Limited Integration of Text Normalization with ABSA:** Text normalization is commonly treated as a separate preprocessing stage rather than being systematically integrated into the ABSA framework. A more effective approach is required

to preserve semantic information while reducing lexical variations in noisy Hinglish text.

- **Limited Aspect-Aware Contextual Modeling:** Although BiLSTM, attention mechanisms, and transformer-based models have demonstrated strong performance in ABSA, limited work specifically investigates aspect-aware contextual modeling for noisy and code-mixed Hinglish reviews [12, 8].
- **Limitations of Word-Level Representations:** Word-level representations may struggle with unseen, rare, and misspelled Romanized Hindi words. Subword-aware representations can provide better robustness by capturing meaningful character-level patterns and reducing the impact of vocabulary variation.
- **Limited Evaluation on Natural Hinglish E-commerce Data:** Existing Hinglish ABSA datasets and studies provide useful benchmarks, but there remains a need for systematic evaluation on naturally collected and manually annotated Hinglish e-commerce reviews. This motivates the development and evaluation of a transliteration-aware, aspect-sensitive model for real-world review data.

3.2 Sequential Processing Pipeline in Existing Systems

Existing sentiment analysis systems generally follow a pipeline structure[3]:

Raw Text → Cleaning → Tokenization → Embedding → Sequence Modeling → Classification

3.2.1 Raw Text

Raw text refers to unprocessed natural language data collected from sources such as social media, online reviews, reviews, or documents. It often contains noise such as punctuation inconsistencies, emojis, and irrelevant symbols. This stage represents the original input before any transformation. It serves as the foundation for all subsequent processing steps.

3.2.2 Cleaning

Cleaning involves preprocessing the raw text to remove noise and irrelevant elements. This may include removing URLs, special characters, stopwords, and correcting basic formatting issues. The goal is to standardize the text while preserving meaningful linguistic content.

3.2.3 Tokenization

Tokenization is the process of splitting text into smaller units such as words, subwords, or characters. These tokens form the basic building blocks for

further processing. Proper tokenization ensures that semantic meaning is preserved. It directly affects the performance of embedding and modeling stages.

3.2.4 Embedding

Embedding converts tokens into dense numerical vector representations. These vectors capture semantic and syntactic relationships between words. Techniques such as Word2Vec, GloVe, or contextual embeddings like BERT are commonly used. This representation allows machine learning models to process textual data.

3.2.5 Sequence Modeling

Sequence modeling captures contextual relationships between tokens in a sentence. Models such as RNNs, LSTMs, or Transformers are typically used for this purpose. It helps in understanding dependencies across different parts of the text.

3.2.6 Classification

Classification is the final stage where learned representations are mapped to output labels. Common outputs include sentiment classes such as positive, negative, or neutral. In advanced systems, classification may also be aspect-specific. This step produces the final prediction of the model.

3.3 Analysis Techniques in Literature

Lexicon-Based Methods: Early sentiment analysis approaches rely on sentiment dictionaries where words are assigned predefined polarity scores. While simple and interpretable, these methods fail in contextual understanding, especially in code-mixed and sarcastic text [3].

Machine Learning-Based Methods: Traditional models such as Naive Bayes, SVM, and Logistic Regression use handcrafted features like TF-IDF and n-grams. These approaches improve performance over lexicon-based systems but depend heavily on feature engineering and struggle with semantic complexity [3].

Deep Learning-Based Methods: Recent studies indicate that neural network architectures such as CNNs, RNNs, and LSTMs significantly enhance sentiment classification performance by effectively modeling sequential dependencies in text. More advanced approaches, including attention mechanisms and transformer-based models, further improve results by prioritizing contextually important words [10, 11, 5, 19]. In particular, BiLSTM with attention has demonstrated strong effectiveness in capturing bidirectional contextual information while emphasizing sentiment-relevant tokens.

3.4 Aspect-Based Sentiment Analysis

Aspect-Based Sentiment Analysis (ABSA) is a fine-grained sentiment analysis approach that identifies sentiments expressed toward specific aspects or features within a text, rather than assigning a single sentiment label to the entire sentence. Each aspect refers to a particular component, attribute, or entity mentioned in the input. This allows the model to capture multiple sentiments within the same sentence [20].

For example, in the sentence: *"The camera quality of this phone is excellent, but the battery life is poor."* ABSA identifies two aspects: **camera quality** (positive sentiment) and **battery life** (negative sentiment).

Compared to traditional sentiment analysis, which assigns a single overall label, ABSA provides more detailed and informative insights. This makes it more suitable for applications such as product reviews and customer feedback analysis, where users express mixed opinions about different features within the same text.

3.5 Transliteration and Normalization

Transliteration refers to the process of converting words from one script to another based on phonetic equivalence. In practical NLP applications, especially in user-generated text, it is closely related to phonetic normalization, where different spellings of the same word are mapped to a common representation [2].

For example, the sentiment-bearing word *"good"* may appear in various informal forms such as *"goood"*, *"gud"*, or *"guddd"*. Although these variations differ in spelling, they convey the same semantic meaning and sentiment intensity. Such inconsistencies are common in social media, online reviews and review data.

Normalizing these variations is important because it reduces vocabulary sparsity and improves model generalization. By mapping different phonetic variations to a standard form, the model can better learn sentiment patterns and make more accurate predictions in downstream tasks.

3.6 BiLSTM and Comparison with LSTM

Long Short-Term Memory (LSTM) networks are a type of recurrent neural network designed to capture long-term dependencies in sequential data. LSTMs process text in a single direction (typically left to right), using memory cells and gating mechanisms to retain or forget information over time. This makes them effective for modeling context in sentiment analysis tasks [4, 11].

However, LSTM considers only past context, which may limit its ability to fully understand sentence meaning. For example, in the sentence *"The movie was not good at all"*, the word *"good"* appears positive

in isolation, but the preceding word *"not"* reverses the sentiment. A unidirectional LSTM may not fully capture such bidirectional dependencies effectively.

Bidirectional LSTM (BiLSTM) addresses this limitation by processing the sequence in both forward and backward directions. This allows the model to capture both past and future context for each word. As a result, BiLSTM provides a richer representation of the sentence.

For instance, in the sentence *"The camera is good but the battery is bad"*, BiLSTM can simultaneously learn that *"good"* is associated with *"camera"* and *"bad"* is associated with *"battery"*, leading to more accurate sentiment interpretation.

Overall, BiLSTM generally outperforms standard LSTM in sentiment analysis tasks due to its ability to incorporate bidirectional contextual information, making it more suitable for complex language understanding problems.

3.7 Attention Mechanism in BiLSTM

The attention mechanism allows the model to focus on the most relevant parts of the input text when making predictions. Instead of treating all hidden states equally, attention assigns different importance weights to different words based on their contribution to the final sentiment [21, 5].

In a BiLSTM-Attention architecture, the BiLSTM first processes the input sequence in both forward and backward directions to generate contextual hidden states. The attention layer then computes a weighted sum of these hidden states, where higher weights are assigned to sentiment-relevant words.

For example, in the sentence *"The phone is good but the battery is very poor"*, words like *"good"* and *"poor"* carry stronger sentiment signals compared to neutral words such as *"the"* or *"is"*. The attention mechanism helps the model focus more on these important words while reducing the influence of less relevant tokens.

Compared to standard BiLSTM models, BiLSTM with attention provides better interpretability and improved performance, as it explicitly identifies which words contribute most to the final classification decision. This makes it particularly effective for sentiment analysis tasks where key opinion words determine the overall meaning.

3.8 Dataset

Code-Mixed datasets used for sentiment analysis are typically collected from platforms such as Twitter, online review sites, and discussion forums [6, 7]. These datasets contain large volumes of user-generated content that reflect opinions, emotions, and feedback across various domains.

Such datasets are inherently noisy and unstructured, often including informal language, spelling variations, abbreviations, emojis, and irrelevant symbols.

This makes preprocessing an essential step before applying machine learning or deep learning models. In addition, the presence of multilingual and code-mixed text increases the complexity of data handling.

For this work, the datasets consist of Hinglish (Hindi-English code-mixed) text collected from e-commerce product reviews. The data introduces additional challenges such as transliterated words, inconsistent spellings, informal expressions, and frequent language switching within sentences. These characteristics require specialized preprocessing techniques to ensure a consistent representation of the textual data.

Dataset 1: Meesho Product Reviews: The first dataset consists of 1,010 Hinglish reviews collected from Meesho product reviews. The dataset primarily contains informal and noisy text, spelling variations, slang, and e-commerce domain-specific vocabulary. The reviews are aspect-rich and opinion-oriented, making them suitable for aspect-based sentiment analysis. This dataset is used as the primary dataset for developing and evaluating the proposed ABSA framework.

Dataset 2: Flipkart Mobile Reviews: The second dataset consists of 128 Hinglish reviews collected from Flipkart mobile product reviews. The dataset contains both Devanagari and Roman-script text, along with informal expressions, spelling variations, and slang. The reviews are manually annotated into three sentiment categories: Positive, Negative, and Neutral. This dataset is currently being developed and expanded to provide an additional naturally collected Hinglish e-commerce dataset for evaluating aspect-based sentiment analysis.

Overall, the objective of constructing these datasets is to develop naturally collected Hinglish e-commerce data that captures the linguistic variations and characteristics of real-world user reviews, thereby providing a suitable basis for studying transliteration-aware and aspect-based sentiment analysis.

3.9 Model Evaluation

Deep learning models such as BiLSTM, attention-based networks, and transformer-based architectures are widely used for sentiment analysis. Their performance is commonly evaluated using metrics such as accuracy, precision, recall, and F1-score. The proposed model will be evaluated using these standard metrics to assess its effectiveness in aspect-level sentiment classification.

The evaluation results reported in the literature provide useful benchmarks for understanding the performance of different approaches on code-mixed and e-commerce review datasets. Table 1 summarizes the approaches and reported results of the six studies reviewed in this work.

The evaluation metrics used in this work are defined as follows:

- **Accuracy:** Measures the proportion of correctly classified instances among all instances.
- **Precision:** Measures the proportion of correctly predicted instances among all instances predicted as a particular class.
- **Recall:** Measures the proportion of correctly identified instances among all actual instances of a particular class.
- **F1-score:** Provides a balanced measure of precision and recall and is particularly useful when class distributions are uneven.
- **Confusion Matrix:** Provides a class-wise breakdown of correct and incorrect predictions.

These metrics will be used to evaluate the proposed model across the positive, negative, and neutral sentiment classes. The results reported by existing studies serve as reference benchmarks; however, direct comparison should be made cautiously because the studies use different datasets, tasks, evaluation settings, and metrics.

3.10 Work Done So Far

The work completed so far includes literature review, dataset creation and preprocessing, and the implementation of benchmark models. A systematic review of existing research on ABSA, code-mixed sentiment analysis, Hinglish text processing, and deep learning-based approaches was conducted to identify existing methods and research gaps.

Dataset Creation and Preprocessing: E-commerce product reviews were collected from Meesho and Flipkart and manually annotated for aspect-sentiment information. The collected data was subjected to a sequence of preprocessing operations, including repeated character reduction, emoji and symbol removal, text normalization, transliteration, and translation. These steps were performed to reduce the lexical and linguistic variability present in naturally occurring Hinglish reviews.

Dictionary-Based Normalization and Unknown Words: A dictionary-based approach was developed to handle variations in Romanized Hindi words. A normalization dictionary is used to map different spellings and variants of a word to a common normalized form. For example, variations such as “acha”, “achha”, and “accha” can be mapped to a consistent representation. This reduces vocabulary fragmentation caused by inconsistent Romanized spellings.

Words that cannot be matched with the existing dictionary are identified as unknown words and stored separately in an unknown-word file. These words are retained rather than being discarded, allowing their occurrence to be analyzed and the normalization dictionary to be expanded iteratively. This approach

Table 1: Comparison of Existing Models for Sentiment Analysis and ABSA

Paper	Model / Approach	Dataset	Reported Result
Aspect-Based Sentiment Analysis and Emotion Detection for Code-Mixed Review [13]	Random Forest, Extra Trees, SVM, GRU, BiLSTM	Code-mixed restaurant reviews	F1: 92% (ET)
Sentiment Analysis in Code-Mixed Telugu-English Text with Unsupervised Data Normalization [14]	Unsupervised normalization + MLP	Telugu-English social text	Accuracy: 80.22%
C1 at SemEval-2020 Task 9: SentiMix: Sentiment Analysis for Code-Mixed Social Media Text using Feature Engineering [15]	CNN + Self-Attention LSTM	SentiMix Hindi-English tweets	F1: 67.1%
SemEval-2020 Task 9: Overview of Sentiment Analysis of Code-Mixed Tweets [16]	BERT-based / Ensemble	SentiMix Hindi-English tweets	F1: 75.0%
Aspect-based sentiment classification of user reviews to understand customer satisfaction of e-commerce platforms [17]	LSTM, BERT, DistilBERT, RoBERTa, XLNet	Trustpilot reviews	F1: 92.53%
Quality achhi hai (is good), satisfied! Towards aspect based sentiment analysis in code-mixed language [18]	Deep learning + monolingual / multilingual embeddings	ABSA-Mix Hinglish reviews	Accuracy: 74.02%

is particularly useful for naturally collected Hinglish data, where new transliterated words, spelling variations, slang, and domain-specific terms may not be covered by a predefined dictionary.

Benchmark Generation: Benchmark models reported in existing research were implemented and evaluated on the collected dataset. These experiments provide baseline results against which the proposed transliteration-aware BiLSTM-Attention model can subsequently be compared.

3.11 Summary of Research Direction

Existing research in Aspect-Based Sentiment Analysis has achieved significant progress through deep learning approaches, particularly using BiLSTM networks and attention-based mechanisms, which effectively capture contextual and aspect-level sentiment information. These models have demonstrated strong performance in standard monolingual settings, especially when supported by advanced embedding techniques and hybrid architectures.

However, most existing studies primarily focus on well-formed textual data and do not adequately address the challenges posed by code-mixed language environments. In particular, Hinglish text introduces additional complexity due to transliteration variability, spelling inconsistencies, and informal writing patterns, which often degrade model performance.

This gap motivates the need for a more robust approach that can handle linguistic noise and variability. The proposed Transliteration-Aware BiLSTM-Attention framework aims to address these limitations by incorporating normalization techniques, subword-level embeddings, and aspect-focused attention, thereby improving sentiment understanding in noisy multilingual and code-mixed scenarios.

4 Conclusion

Sentiment analysis using Natural Language Processing (NLP) has become a significant research domain due to the exponential growth of social media platforms and the continuous generation of user-generated content. These platforms serve as rich sources of public opinions, emotions, and feedback related to diverse domains such as products, services, politics, and social issues. Consequently, sentiment analysis plays a crucial role in enabling organizations, researchers, and policymakers to extract meaningful insights from large-scale textual data.

Early sentiment analysis systems were primarily based on lexicon-based methods and traditional machine learning techniques such as Naive Bayes, Support Vector Machines, and decision trees. While these approaches performed reasonably well on structured data, they were limited in capturing contextual de-

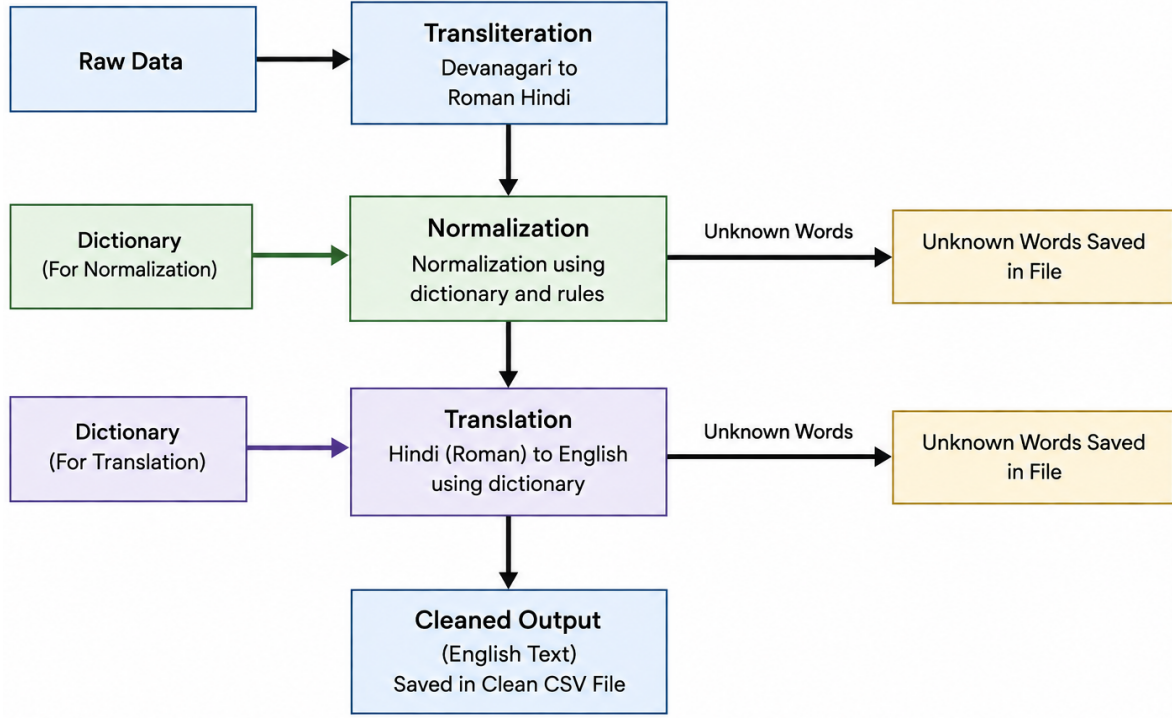


Figure 2: Data Preprocessing Flow

dependencies and complex linguistic patterns.

With advancements in deep learning, modern approaches leveraging recurrent neural networks, BiLSTM, and transformer-based architectures have significantly improved the ability to model contextual and semantic relationships in text [1, 10]. These models have enhanced performance in tasks such as opinion mining, emotion detection, and aspect-based sentiment analysis. However, challenges such as informal language usage, sarcasm, multilingual text, and noisy data continue to impact model reliability and accuracy [8, 9].

In conclusion, while substantial progress has been made in sentiment analysis, further improvements are required to handle real-world linguistic complexity more effectively. Future research should focus on developing more robust and context-aware models capable of handling multilingual communication, sarcasm, and informal text patterns. The integration of advanced preprocessing techniques, contextual reasoning, and multimodal learning is expected to further enhance the performance and applicability of sentiment analysis systems in real-world scenarios [7].

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